

Divided Democracy: Analyzing the Impact of Stricter Voting Laws on Party-Specific Voter Accessibility

DATASCI 203: Lab #1

Maria Lee, Jackie Loyola and Tracy Volz

Contents

1	Importance and Context	1
2	Data and Methodology	1
3	Results	3
4	Discussion	3

1 Importance and Context

After the controversial 2020 election, the American political landscape was deeply divided. The division over the election’s legitimacy led to a stark contrast between the Republicans’ and Democrats’ views on voting laws. Republicans took a stance that stricter voting laws were necessary to protect the integrity of elections and to ensure that only eligible voters had their voices heard. Many Republican-controlled state legislatures moved to pass stricter measures, such as limiting the ease of mail-in voting or requiring specific voter identification to prevent voter fraud.

On the other side, Democrats opposed these changes to voting laws. They argued that these stricter measures were unnecessary and that they would disproportionately affect minority and low-income communities, which often faced more challenges in obtaining the required identification. Democrats believed that these voting laws were an attempt to make it harder for Democratic voters to participate in elections.¹ *Rock the Vote*, a non-profit American organization that encourages young voters to participate in the election process, has hired us to perform a study to analyze the effects of the stricter voting measures. As a first step in this analysis, this research aims to address the following question:

Do Democratic voters or Republican voters experience more difficulty voting?

By answering this question, we can potentially provide insight into whether certain political parties are experiencing more difficulty voting since the implementation of stricter voting laws from the 2020 election. Our client’s goals are to spread awareness of the controversial American voting process and to provide guidance to policymakers to hopefully find a reasonable balance on voting laws between the two major parties.

2 Data and Methodology

Our analysis utilizes information sourced from the American National Election Studies (ANES). This observational pilot study was conducted online by YouGov between November 14, 2022, and November 22, 2022. The purpose of this pilot study was to test new questions for ANES’s 2024 Time Series Study and to provide information regarding voting and public opinion after the 2022 US midterm elections. The dataset consists of 1,585 respondents, of which 1,500 were weighted to accurately represent the U.S. population after data collection, considering factors such as the 2020 presidential candidate choice, gender, age, education, and race. This analysis did not incorporate the weights because we cannot perform mathematical operations on ordinal data. Participants were rewarded for their participation in completing the questionnaire.²

Each row of the dataset represents responses from one participant. For the sake of research accuracy, it is important to precisely define the terms used in this specific analysis. In determining whether a voter leans towards ‘Democrat’ or ‘Republican’ affiliation, we calculated the number of times a voter supported a respective party candidate in the 2022 U.S House, Senate, Governor elections, and the 2020 Presidential election (ANES codes: house22p, senate22p, gov22p, and vote20). For individuals who were unable to cast a vote, the general question of party identification was used to determine their affiliation (ANES codes: pid1d, pid1r, and pidlean). If a voter’s support for Democratic party identification or candidate selection exceeded that of the Republican party, they were categorized as ‘Democrat,’ and vice versa. The results are captured in a derived column called ‘party_affiliation.’ With this report, we aim to assess variations, if any, in voting challenges between Republicans and Democrats. Our analysis excluded all respondents categorized as ‘Other.’ In Table 1, we report that 662 respondents are classified as Democrats (~42%), 631 were classified as Republicans (~40%), and 292 were classified as ‘Other’ (~18%).

A ‘voter’ is defined as an individual who meets the following qualifications to participate in the election process: they must be a U.S. citizen, at least 18 years old, and have responded to the questionnaire regarding casting a vote. Our research primarily focuses on individuals who responded to this question, as challenges related to voting could have hindered their ability to vote. Excluding them solely based on their non-participation

¹Harte, J., Trainor, C. (2022, November 1). Where voting has become more difficult. Reuters. <https://www.reuters.com/graphics/USA-ELECTION/VOTING-RESTRICTIONS/zvnbdbjbkvl/index.html>

²2022 Pilot Study - ANES: American National Election Studies. ANES | American National Election Studies. (2022, December 14). <https://electionstudies.org/data-center/2022-pilot-study/>

would introduce survivorship bias into our study. Regardless of whether or not a participant cast a vote, if they were classified as a Republican or a Democrat and they answered the following question, they were included in our analysis:

How challenging was it for you to vote?

In the context of our study, the term ‘challenge’ or ‘difficulty’ is defined as a subjective measure of the amount of effort needed to successfully cast a vote in the 2022 election. The data is reported on a 5-point Likert scale that ranges from 1 (‘Not very difficult’) to 5 (‘Very difficult’). Out of the 662 Democrats, 529 provided responses to this question (~88%). Out of 631 Republicans, 581 respondents answered this question (~84%). Only participants who answered this question were included in our analysis. The dataset was complete, and there were no missing values. Refer to Table 1 for the final counts and proportions.

Table 1 - Summary of Data

Party Affiliation	Classified (Count)	Classified (Proportion)	Answered “Difficulty” Question (Count)	Answered “Difficulty” Question (Proportion)
Democrat	662	0.42	581	0.88
Other	292	0.18	NA	NA
Republican	631	0.40	529	0.84

Bar charts for each group (shown in Figure 1) demonstrate a comparable distribution shape, which suggests the groups are drawn from populations with similar underlying characteristics.

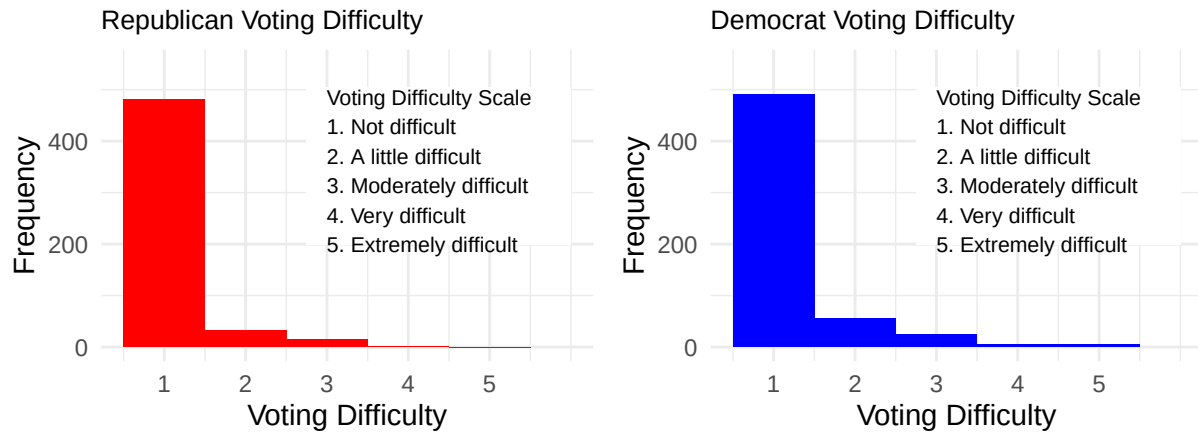


Figure 1: Bar charts of Voting Difficulty for Republicans and Democrats

In an ideal scenario for comparing two groups, the t-test is preferred when working with normally distributed metric data, as it provides high statistical power and accuracy. However, our study involves ordinal data. Since our goal is to compare two distinct and mutually independent groups of ordinal data, the most suitable approach is to conduct a ‘Comparison of Hypotheses’ version of the Mann-Whitney U Test (also known as the Wilcoxon Rank Sum test).³

The null hypothesis of our Mann-Whitney U Test can be phrased as follows:

The probability a Democrat had more difficulty than a Republican is equal to the probability a Republican had more difficulty than a Democrat

The assumptions for the Mann-Whitney U Test are as follows:

- Data must be identically distributed - The ANES 2018 pilot study relies on a panel of individuals recruited from the YouGov platform, potentially leading to the presence of dependencies. For example,

³R Documentation: Mann-Whitney U Test (n.d) <https://search.r-project.org/CRAN/refmans/sjstats/html/mwu.html>

geographical clustering may have occurred in regions with unfavorable weather conditions, impacting the mood of respondents with similar political affiliations during the survey or voting process. While there are several potential scenarios that could have deviated from strict Independence and Identically Distributed (IID) assumptions, the available data does not show distinct violations of the IID principles.

- Data must be ordinal or better - Our data is ordinal.
- Each Democratic and Republican sample are separately drawn from the same distribution and all are mutually independent - Our grouping criteria allows a person to be either a Democrat or a Republican (not both). The bar charts demonstrate that the groups follow a similar distribution.

All Mann-Whitney U Test assumptions are met.

3 Results

```
mannwhitney(data = subset_data_votehard, x = votehard, grp = party_affiliation,
             distribution = "asymptotic",)
```

```
##
## # Mann-Whitney-U-Test
##
## Groups 1 = Democrat (n = 581) | 2 = Republican (n = 529):
##   U = 332775.000, W = 163704.000, p = 0.001, Z = 3.263
##   effect-size r = 0.098
##   rank-mean(1) = 572.76
##   rank-mean(2) = 536.54
```

The results of the Mann-Whitney U comparison test provide evidence suggesting that Democrats encounter more voting difficulties than Republicans (p-value = 0.001). Under standard testing framework, we reject the null hypothesis. However, the calculated effect size (r) produced was equal to 0.098. The effect size (r) quantifies the magnitude of the observed difference, helping to assess the practical significance of the results in a statistical analysis. Commonly accepted interpretations categorize effect sizes as “small effect” (0.1 - 0.3), “medium effect” (0.3 - 0.5), or “large effect” (> 0.5) to assess practical significance in statistical analysis ⁴ Our measured effect size was lower than 0.1, therefore we conclude these results yield no practical significance for observed differences between Democrats and Republicans.

Our analysis has a limitation that affects the inferences we can draw. As mentioned above, we were not able to incorporate ANES weights into our analysis because our survey question was of an ordinal scale. According to Mathew DeBell, Director of Stanford operations for the ANES, it is advisable to utilize the weights “because unweighted ANES data do not represent the population as well as weighted data.” ⁵

4 Discussion

For the 2022 election, we found no practical significance in the observed differences between Democrats and Republicans. As a follow-up analysis, we aim to integrate the ANES dataset with participants’ demographic information. This will allow us to explore the combined pre-categorized difficulty responses and identify patterns related to party affiliation. The aggregated data is considered metric, enabling us to apply weighting criteria for each participant, which will enhance the accuracy of the results. Furthermore, this information will contribute to *Rock the Vote*’s mission by helping determine whether young voters in either affiliation encounter difficulties when casting their votes.

⁴R Documentation: Wilcoxon Effect Size (n.d) https://search.r-project.org/CRAN/refmans/rstatix/html/wilcox_effsize.html

⁵DeBell, M. (2010, May). How to Analyze ANES Survey Data . ANES | American National Election Studies . <chrome-extension://efaidnbmnnnibpcajpcgclefindmkaj/https://electionstudies.org/wp-content/uploads/2018/05/HowToAnalyzeANESData.pdf>